Web technology

The tools, knowlege & procedure (Algorithms associated with world wide web is consider as web technology. It is usefull for developing running & upgrading woold wide web

* Note on:
Inteonet, intranet, extranet, www webside webpage

Markup language → (Guide broweser how to display content by using tag)

It is not a program language, it list arrange the content & guide broweser how to display content in a web page. It uses tags to mark the content
commonly used markup language are HTML XML SGML (standerdize) WML (wireless) CML (extensible) (chemical)

HTML
It is a markup language that is used to display content in a webpage in attractive and informative way. It uses predefineeed tages to mark the content & guide broweser how to display content in a webpage

Tags → Tages are predefined words in closed angle bracket (<>). Every tag has

predifine meaning & procedure to use eg;
<H1> - - - - - </H>, <p> - - - - </p> etc.

Type of tag
i singular tag / Empty tag
ii paired tag / non-empty tag container tag

i singular tag
The tag without closing tag is called singular tag. This tags don't contain any ~~containce~~ content But its effect is seen in a webpage eg </Hr/>

, <img/>, <input/>

ii) paired tag
when a tag contains within it's tag it can contain plain text other tags as a child element with in its opeaning & closing tags

Basic Structure of HTML

<HTML>
<HEAD> } Head part
</HEAD>
<BODY> } body part
</BODY>
</HTML>

every webpage must follow basic structure of HTML Basic structure contains root tag HTML It contains entire content of webpage with in opening and closing HTML Tag. It divided into two part Head part & Body part.

Headpart contains info related to web page it is not visiable in a webpage but its effect is seen It is enclosed in opeaning & closing head tag. It is generally contains tittle tag meta tag ~~ting~~ link tag script tag style tag etc.

Title tag: It is use to display title of ~~related of~~ ~~webpage that make visible to~~ sear webpag in a title bar

meta tag: meta tag is used to insert keywords related to webpage that make visible to search engine.

Link tag: It is used to link external File to the web page link css & Js file.

script tag: It is used to insert Js code in webpage

body tag

Body part ~~It~~ contains all the visible contents of the webpage within opeaning & closing body tag. It generally contains heading tags, paragraph tags Formalating tage Image tags, list tags table tag, form tag frame tags misc. etc tags

Attribute
It is the additional info provided to tag that is used to specifi any tag every tags has predifined set of attributes & their respective values.

The tags used in HTML

-> Body tag

• It contains all the visible content the web page. It can be used only once in a webpage

Syntax — Attribute

<Body> — bg color = "Red" | "#f00"

:

</BODY> — background = "abc.jpg" | "abc.png" | "abc.gif" | "abc.bmp"

-> Heading tag

Heading tag are used to display titles of the content. There are 6 level of hedding tags

Syntax — Attribute

<H1> ---- </H1> — align = "left" | "Right" | "center"

<H6> ---- </H6>

-> Paragraph tag

It is used to display content in paragraph format

Syntax : <p> .... </p>

attribute: align = "left" | "Right" | "center" | "justify"

Note : paragraph tag can contain other tag as an child element

Formatting tag
They are used to decorate content webpage there are several formatting tags used in webpag. Some common tags are:

<u> ---- </u> underline
<B> ---- </B> Bold
<I> ---- </I> Italic
⁻⁻⁻⁻ super script
₋₋₋₋ subscript
<Font> ---- </Font>

Attribute
size = "18pt"
color = "Red"
style = "monotype corsiva"

<HTML>

<HEAD>

<title> web Technology </Title>

</HEAD>

<BODY>

<H1>

<P> - - - - - - - - - - </P>

<B> computer </B>

<U> computer </U>

<I> computer </I>

_{H_2O, H_2SO_4}

^{$(a+b)^2$, $(a^2+2ab+b^2)$}

list tag

It is used to display content in proper order Either using alph numerical points or bullets point There are 3 list use in HTML

1 order list

2 unorder list

3 Definition list

1 order list

order list is used to display content in alph numeric order

syntax

<OL>

<LI> - - </LI>

:

:

</OL>

Attribute

type: = "1" / "a" / "A" / "i" / "I"

start "3"

for eg. Number System

a) Binary number

b) octal number

c) Decimal number

d) Hexadecimal number

```
<HTML>
<HEAD> - - - </HEAD>
<BODY>
<H1> Number system </H1>
<OL type = "a">
<Li> Binary number </Li>
<Li> octal number </Li>
<Li> Decimal number </Li>
<Li> Haxa Decimal number </Li>
</OL>
</BODY>
</HTML>
```

ii) Unorder list

It is used to display content in bullet format

syntax: <UL>

<Li> - - - </Li>

:

</UL>

Attribute: type: "Disc" | "Square" | "circle"

eg: Number system

- Binary Number
- octal Number
- Decimal Number
- Haxadecimal Number

```
<HTML>
<HEAD> - - </HEAD>
<Body>
<H1> Number system </H1>
<UL type = "square">
<Li> Binary number </Li>
<Li> Octal number </Li>
<Li> Decimal number </Li>
<Li> Haxadecimal number </Li>
</UL>
</Body>
</HTML>
```

Q.no1 Output

Type of computer

A. micro computer

• portable

1. laptop
2. Not book
3. Tablet
4. palmtop
5. PDA

• Non portable

1. work station
2. personal computer

B. mini computer

C. main frame computer

D. super computer.

<HTML>
<HEAD> - - . . . . </HEAD>
<Body>
<H1> Type of computer </H1>
<ol type = "A">
<LI> micro computer </LI>
<ul type = "circle">
<Li> portable </LI>
<ol type = "1">
 lap top
<Li> Not Book </Li>
<Li> Tablet </Li>
<Li> plamtop </Li>
 PDA </Li>

 non protable
<oL type = "1">
<Li> work statim </Li>
 personal computer </Li>
</OL>
</Ul>
 mini computer </Li>
 main frame computer </Li>
 super computer </Li>
</OL>
</Body>
</HTML>

Q.no Type of memory

I primary memory

• RAM
  a. SRAM
  b. DRAM
• Rom
  a. PROM
  b. EPROM
  c. EEPROM
• Cache
  a L1 Cache
  b L2 cache

II Secondary memory

• magnetics memory
  a. magnetics Disk
    ▪ Hard Disk
    ▪ Flopy Disk
  b. magnetic Drum
  c. magnetic tape
• optical memory
  a. CD
  b. DVD
  c. Blu Ray Disk
• flash memory
  a pen drive
  b memory card
  c solid state drive.

```
<HTML>
<HEAD> - - - - </HEAD>
<BODY>
<H1> type of memory </H1>
<ol type = "I">
<Li pramery memory </li>
  <ul type = "circle" </ul>
    <li> RAM </Li>
      <ol type = "a">
        <Li> SRAM </Li>
        <Li> DRAM </Li>
      </ol>
    <li> ROM </Li>
      <ol type = "a">
        <li> PROM </li>
        <li> EPROM </Li>
        <Li> EEPROM </Li>
      </OL>
    <Li> cache </Li>
      <ol type "a">
        <li> L1 cache </Li>
        <li> L2 cache </Li>
        <lol>
    <lul>
    <li> secondary memory <li>   <ul type circle>
      <ol type = "a">
      <li> magnetic Disk <li>
      <ul type = "square">
      <li> Hard disk </li>
<li> floppy disk </Li>
<lul>
<li> magnetic drump </le>
<li> magnetic tape </Li>
</OL>
<li> optical memory <li>
 <ol type = "a">
 <li> CD </li>
 <li> DVD </li>
 <li> Bluray Disk </Li>
 </ul>
 <Di > flash memory </Li>
 <ol type = "a">
 <li> pendrive </li>
 <li> memory card </Li>
 <li> solid state drive </li>
 </ol>
 </ul>
</ol>
</Body>
</HTML>
```

Hyperlink

Hyperlink is an HTML object that is used to link two object with documents or two websits two web document It used an char at a to insert hyperlink in a webpage
Hyperlink can be inserted on the next images or objects.

syntax

<a> ...

Attribut

name = "abc"

href = "#abc"|"abc.html"|
"www.abc.com"

target = "_blank"|"_self"|"_parent"
|"_child"|name of frem

Types of Hyperlink

Internal Hyperlink

local Hyperlink

extrenal Hyperlink

Internal Hyperlink

When two ancher objects are linked within the webpage then it is caled internal hyperlink. In this hyperlink one anchor tag is used for book marking and another anchor tag is used for refrencing.

Example

Number System

1. Binary number
2. Octal Number
3. Decimal Number
4. Hexa decimal Number

Binary Number

Top

octal Number

Top

Decimal Number

Top

Haxadecimal Number

Soln
```
<HTML>
<Head> ..... </Head>
<Body>
<H1> <a nam = "top">Namber system </a> </H1>
<OL>
<Li><a href = "#Binary">Binary Number </a></Li>
<li><a href = "#octal">octal Number </a></li>
<li><a href = "# decimal">Decimal Number</a></li>
<li><a href = "#Hexdecimal">Haxdecimal Number</a></li>
</ol>
<H1><a name = "binary">Binary Number </a></H1>
<p allgn = "Justify">
```

</P>
top
<H2> octalNumber</H2>
<p align = "justfiy">

<p>
top
<H2> DecimalNumber</H2>
<p align = "justfy">

</P>
top
<H2><a name = "Hexadecimal") HexdecimalNumber</H2>
<p align = "justfy">

</P>

Local Hyperlink

When two web document are link together with the help of anchor tag within the web side then it is called local Hyperling in this hyperlink we used name of the document for reference.

eg.
 ABC

external Hyperlink

When two webside are link with the help of anchor tag then it is called external hyperlink in this hyperlink URL is used for referencing

eg.
Google

table tag

Table tage is used to insert table in webpag it arrange contane tebular structure that is in row & column structure it is also used to layout web page

Syntax
<table>
<TR>
<TD>.... </TD> TD>.... </TD>
</TR>

</Table>

TR = Table row
TD = Table data
TH : Table Header

~~For discrete & continuous series~~

~~Df F~~

1	2	3
4	5	6
7	8	9

```
<HTML>
<HEAD>... </HEAD>
<Body>
<Table border = "1">
<TR>
<TD> 1 </TD> <TD> 2 </TD> <TD> 3 </TD>
</TR>
<TR>
<TD> 4 </TD> <TD> 5 </TD> <TD> 6 </TD>
</TR>
<TR>
<TD> 7 </TD> <TD> 8 </TD> <TD> 9 </TD>
</TR>
</Table>
</Body>
</HTML>
```

Attribut of table tag

align = "center" | "left" | "Right"

height = " 400px"

width = " 400px"

bg color = "Red" / #FF0000"

background = "abc.jpg"

cellspacing = "0px"

cellspacing = "5px"

boader = "2px"

boadercolor = "Red" | "green" | "#FF0000"|

Attribute of table row tag (TR)

bg color = "orang" | "#0000

background = "abc.jpg"

align = "left" "Right" | "center"

valign = "top" | "middle" | "bottom" (V: vertical)

height = "100 px"

width = x no. TR width = table cell

Attribute of TH / TD

heigh = "x"

width = 400 px

bgcolor = "Red" | "#FF0000

background = "abc.JPG"

align = left" | "Right" | "Central"

valign = "top" | "middle" | "bottom"

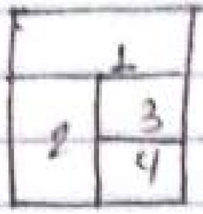
Row span = " "

cols span = " "

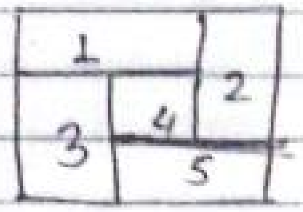
Read	green	Pink
olive	yellow	Brown
orange	purple	gray

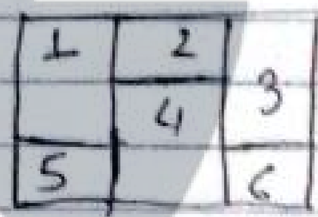
```
<HTML>
<Head> - ---   </Head>
<body>
<Table border = "1">
<TR>
<Td background = "Read" >Read</Td>
<Td background = "green"> green</Td>
<Td background = "pink"> Pink </Td>
</TR>
<TR>
<Td background = "olive" >olive</Td>
<Td background = "yellow" >yellow</Td>
<Td background = "Brown"> Brown</Td>
</TR>
<TR>
<Td background = "orange" > orange</Td>
<Td background = "purple"> purple</Td>
<Td background = "gray"> gray</Td>
</TR>
</Table>
</body>
</HTML>
```

Web span & colspan of table



```
<HTML>
<TR>
<TD Colspan = "2" > 1</TD>
</TD>
<TR>
<TD Rowspan = "2" </TD><TD>3</TD>
</TR>
<TR>
    <TD>4</TD>
</TR>
</Table>
```





```
<HTML>
<Head></Head>
<Body>
<TABLE>
<TR>
<TD colsspan = "2" >1</TD>
<TD Rowspan = "2" >2</TD>
</TR>
<TR>
<TD Rowspan = "2" > 3</TD><TD>4</TD>
</TR>
<TR>
<TD colspan = "2" >5</TD>
</TR>
</TABLE>
```

<TABLES>
<TR>
<TD Rowspan = "2" >1</TD><TD>2</TD>
<TD Rowspan = "2"> 3</TD>
</TR>
<TR>
<TD Rowspan = "2">4 </TD>
</TR>
<TR>
<TD>5</TD> <TD> 6 </TD>
</TR>
</TABLE>

form
form is a HTML object that is use to receive data from user. it is the only object through with user form submit data to the website of recept data with the help of different from element

Syntax
<from>
⋮
<from element>
</from>

attribute
name
method : "get" /"post"
action = "login php"
mailto

form element
form element are components of form through which user input date there are 4 mean form element
I label
ii) select
iii) text area
iv) input

i) label tag:
It is a form element that is used to display caption of other form elements.
syntax: <label> - - - - · · · · </label>
attribute: name =
text =

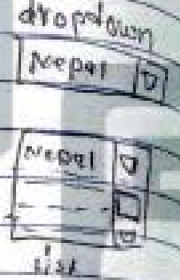
ii) select tag:
It is a form element that is used to insert a combo box / dropdown box & list box. It is followed by option tag as a child element.

Attributes

name =

multiline rows = "1" "5"

multichar cols =



syntax:
<select>
<option>......</option>
</select>

iii) text area
It is a form element that is used to insert multiline text box. It can support large no. of characters.

syntax
<text area>.....</textarea>

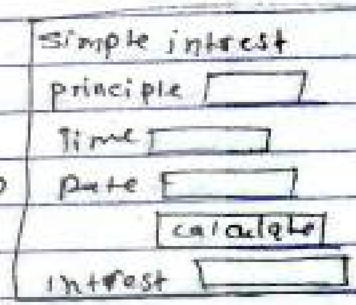
Attributes
name =
rows =
cols =

iv Input
Input is a form element that is used to insert different input obj like text box radio-buttons, buttons etc.

syntax
<input>

attributes
name :
Value =
size =
type = "text [ ] textbox
= "password" [****]
= "radio" ⊙
"checkbox" [✓]
"file" [ ] Browse
"button" [button]
"submit" [submit]
"Reset" [Reset]
"hidden" X

Example: Write a HTML code to design simple interest entry form.



<HTML>
<head></head>
<body><h1> simple interest </h1>
<from>
<label> principle </label>
<input type = "text" />

<lable> Time </lable> <input type = " text ">
<lable> Rate </lable> <input type = " ~~calcu~~ text ">
<input type = "submit" value = " calculate" />
<label> interest </lable> <input type = " text "/>

</form>
</body>
</HTML>

Registration form

```
<html>
<head></head>
<body>
<h1>Registration form</h1>
<form>
<table width="60px">
<tr>
<td><label>name</label></td>
<td><label>
<td>input type="text"/></td>
</tr>
<tr>
<td><label>password</label></td>
<td><input type="password"/></td>
</tr>
<tr>
<td><label>gender</label></td>
<td><input ="radio" name="r"/>male
<input> type="radio" name="r"/>female
</td>
</tr>
</form>
</body>
</html>
```

2081-11-12

Frame

Frame is HTML object that is used to divide tab of webpage into multiple parts. It integrate multiple webpage into single page It was used to design a lay out of webpage but nowadays it is only used as a popup ads.

Syntax

```
<frameset
    <frame-.... />
        :
 </frameset >
```

Attribute of framemeset

```
rows = "30px *, 150px"
cols = "100px, *"
```

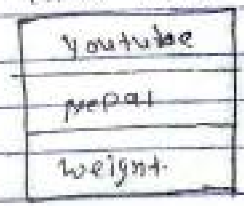
Attribute of frame

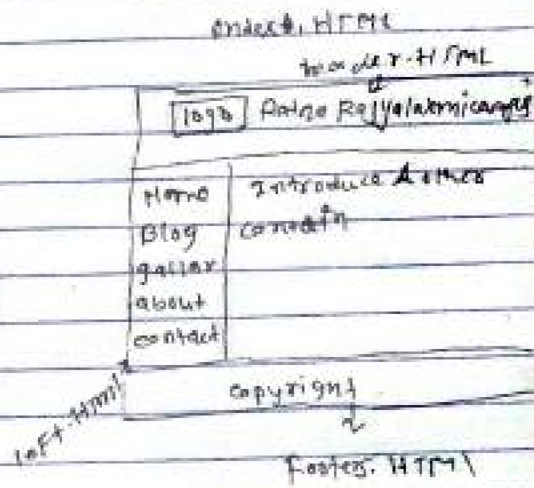
```
name = " "
resizeable = "true" / "false"
border = " "
src = " "
```

Note frame tags are written outside the body tag

```
<HTML>
<head>....</head>
<frameset rows = "33%.*, 33%.">
<frame src = "www.youtube.com"/>
<frame src = "www.Nepal.com"/>
<frame src = "www.weight.com"/>
</frameset>
</HTML>
```

index HTML





Style sheet language

Style sheet language is a seperate language othe than HTML that is used to decorate content of webpage It always works along with markup language (HTML)

The main purpose of scripting language is

1. seperate content from styling.
   Designer can design webpage seperately & indepen dently without thinking of content of webpage.
2. provide additional styling features than attributes of html to decorate content.
3. Implement template concept and design reusability.
   Eg. of style sheet language are css & XSL

Example of style sheet language

1. cascade style sheet language (css)
2. Extensible style sheet language (XSL)

CSS is a style sheet language that is used to decorate content of web page. CSS is widely used style sheet language It provide attractive and dynamic factors to style a web page There are 3 types of css implementation in a web page.

(i) In line style sheet
(ii) Internal style sheet
(iii) External style sheet

(i) Inline Style sheet
When css attributes is define inside the tag of HTML. In this style sheet style keyword is used as an attribute of tag and with the help of this attribute we define css

syntax
<P style = " background: orange; width: 500px">
</P>
Note: The effect of css is applied only to the tag where it is define.

(ii) Internal Style sheet
When css code is written inside the webpage then it is called internal style sheet. In this style sheet style keyword is used as a tag and inside opening & closing style tag we write css code.

generally style tag are placed inside the head part of HTML

```
<HTML>
<Head>
<style>
<body>
{
background: gray;
}
P
{
background: white;
width : 500px;
}
</style>
</Head>
<Body>
<P> - - - </P>
<P> - - - </P>
</Body>
</HTML>
```

Note: In this style sheet the effect of css code is applicable only for this webpage.

(iii) External style sheet
When css code is written in a sperate file with extension css then after the css file is linked with the page with the help of linked tag then it is called external css. In this style sheet the effect of css code is applicable to all the webpages of website $\boxed{\text{index.html}}$

```
<HTML>
<HEAD>
    <link href = "style.css" type = "text/css" rel = "style sheet" />
```

style
CSS

</Head>
<Body>
<p> - - - </p>
</Body>
</HTML>

Sytyle css

```
body
{
background : gray;
}
p
{ back ground: white;
width : 50px;
}
```

CSS selector
css selector is a method to implement css Styling to target tags of HTML. Types of css Selector

1 Universal selector
2 Default selector / tag selecter
3 Id selector
4 class selector
5 Child selector / group selector
6 pseudo selector
7 Attribute selector

1 Universal selector
Universal selector all the tags of HTML and implement css styling to all the tag of HTML. It is denoted by.

```
*{
  // CSS attribute
}
```

eg

```
*{
  margin : 10px;
  padding : 10px;
  back ground ; gray ;
}
```

2 Default selector
Default selector select all the tag which is mention in css. Normally to define default selector we use tag name without angle bracket & implement css style to all the tag.

syntax

```
tag name{
}
```

example

```
p
{
margin : 10px;
padding : 10px;
back ground : gray;
}
```

3 Id selector
Id selector select specific tag where It is mention. Id selector is mention with the help of Id keyword. Id keyword use as an attribute of tag. To define Id selector we use # symbol as a prefix of selector name

Syntax

```
# [Id_name] {

}
```

```
#abc{
  margin
  padding
  background
}
```

<p id = "abc"> - - - </p>

④ class selector
class selector select specific tag where it is mention
class selector is mention with the help of class keyword
use as attribute of tag To define class selector we
use '.' symbol as a prefix of selector name

syntax
```
.[class name]{
    //css attribute
}
```
eg
```
.abc{
    margin: 10px;
    padding: 10px
    background: gray
```

⑤ child selector
child selector is used to select a sub group of tag
It is used to select child element of tag.
```
<div id = "abc">
  <p> - - - </p>
</div>
<div id = "xyz">
<p> - - - </p>
</div>
```
```
#abc p{
    margin: 20px;
    padding: 10px;
    background: olive
}
#xyz p{
    margin
```

⑥ Pseudo selector
pseudo selector is used to implement css style when
some event is occured. It can be applied to any tag the
events are like
① hover
② visited
③ link
eg
```
.{
  width : 500px
}
p : hover
{
  width = 1000px
}
```

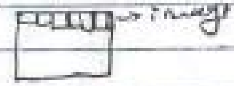
⑦ Attribute selector
It provide a way to target elements more precisely and apply
style based on their attribute It is denoted by square
brackets [ ]

Style.css

```
*{
  margin:0px
  pading:0px;
}
body
{
 background:orange;
}
p
{
 background:while;
 width:800px;
}
#abc
{
 backgrond:read;
 width:500px;
}
.abc
 {
  background:pink;
  width:900px;
 }
 p:hover{
        background:blue;
      }
```

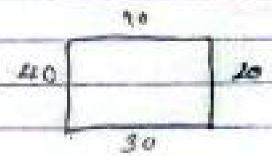
```
<HTML>
<Head><head>
<Body>
<p>--</p>
<p>--</p>
<p>--</p>
<p>--</p>
<p id="abc">---</p>
<p class="abc">---</p>
<p>------</p>
</Body>
<HTML>
```

```
background:red url("abc.jpg") repeat-x 00 fixed;
background-color;
background-image:
background-repeat=repeat-x/repeat-y/no-repeat-y/no-real
background-position x:0
background-position y:0
background-attachment:
```



padding

```
# margin= 10px
# margin= 10px 20px 30px 40px
#margin-top:
 margin-left:
 margin-Right:
 margin-bottom;
```



Web Architecture

Scripting language

Scripting language is similar to programming language that provide programming logic & mathematical calculation in website development. It is used to create a dynamic website that helps to create interactive and attractive user interface design & dynamic content generation

Types of scripting language

1 client scripting language
2 server scripting language

client scripting language
The Scripting language whose program code is translated at computer with the help of browser then it is called client site scripting language. It cannot connect to database so it is difficult to generate content dynamically that's why this scripting language are used to design better user interface. Example. Java script VB script type script

2 Server Scripting language.
The scripting language whose program code is translated at server computer with the help of web server where language translator is integrated. It can connect the data base & executes in web server so it is useful for backend programming eg.- PHP asp/Asp.net JSP CGI script, perl

PHP (Hypertext preprocessor)

php is widely-used open-source server-side scripting language designed for web development It is embedded in HTML and is particularly suited for creating dynamic web pages and applications php scripts are executed on the server and the result is sent to the client's web browser as plain HTML.

Features

opean Sources
php is free to use & distribute. It source code is available for anyone to modify and extend making it highly customizable.

cross-plat form
Php is compatible with various operating systems, including windows linux, macos it also support a wide range of web servers like Apache, IIS & Nginx ~~wide range of web servers like~~

3 Embedded in HTML
php code can be embedded directly within HTML making it easy to mix server-side logic with client-side presentation.

(4) Database Integration
It has built in support for connecting to wide range of database including my sql postgre SQL order & sqlite.

5 Extensive library Support:
php has a rich set of libraries & extensions that provides functionalities for tasks like file handling image processing XML parsing & more.

6 Easy to learn
It has a relatively simple syntax that is easy to understand especially for those with a background in C, Java or perl

7 Error Reporting
php provides comprehensive error reporting which helps developers quickly identify & fix issues in their code.

Different between

programming language	Scripting language
It is used to create stand alone applications & software	It is used to write script for automating tasks or controlling other programs
Runs directly on hardware oros	Runs within a host environment
compiled into machine code	interpreted at runtime
faster due to compilation	slower due to interpretation
It is independent of other programs	It is dependent on a host environment
can directly access system resources	limited access to system resources.
All the programming languages are not scripting language	All the scripting languages are programming language.
It does creates binary files	It does not create any binary files
It is code intensive	It is less code intensive when compared with programming language.

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operator

operators are predefined symbol that perform operations on operand. Operand May be variable constant or expressions

Types of operators
on the basic of number of operand use
1 unary operator (one operand) (++ --)
2 Binary operator (two operand) (+, -, /
3 Ternary operator (three operand) (?)

on the basis of Functionality
1 Arithmetic (+, -, *, /, %)
2 Relational (< <= >, >= ==, !=) ===
3 Logical (&&, ||, !)
4 Increment / Decrement (++, --)
5 assignment (= +=, -=, */ /= , %=)
6 Bitwise &, |! << >> ^)
7 conditional (? :)
8 concatenation (.)
~~concatenation (=)~~

Control structure

control structure is used to control execution of program flow. It guide how the program codes execute.

Types of control structure
1 sequential
2. Conditional structure
3. Loop structure.

Interest

1 Sequential / structure
when program executes step by step in linear flow without any branching then it is called sequential structure. It generally follows input process & out algorithm.

conditional structure / selection / decision

When sequential flow of program execution branches according to condition provided then it is called conditional structure. It select a block that's why it is called selection structure if-else switch-case.

Write a php program to calculate simple interest where rate = 12% if principle greater then 50000 otherwise rate = 7%. time & principle are given by user

```
<HTML>
<Head> </Head>
<body>
<?php
 $p = 60000
 $T = 5;
 if ($p >= 50000)
 {
 $r = 12;
}
else {
   $r = 7;
}
$I = $P * $T * $r / 100;

echo "Interest = $ I";
 ?>
</Body>
</HTML>
```

Write a php to find profit & loss along value where cost price & selling price are given by user

```
<HTML>
<Head> </Head>
<body>
<?php
 $sp = ~~$sp $sp~~ 50000
 $cp = 30000;
 if ($sp > $cp)
 {
 $P = $sp - $cp;
 echo = "profit = $p";
 }
 else if ($cp > $sp)
 {
  $L = $cp - $sp;
  echo "Loss = $ L";
  }
  else {
   echo "No change";
   }
   ?>
```

```
</Body>
</HTML>
```

Loop Structure

Loop structure is a control structure that is used to repeate a block of program code according to condition provided.

- for-loop
- while-loop
- dowhile-loop
- foreach-loop

```
<HTML>
<Head></head>
<Body>
<?php
$n = 23;
    for ($i = 1 ; $ <10; $i++)
    {
     $a = $i * 23;
    }
  echo "23 x $i = $a";
  }
  ?>
</Body>
</HTML>
```

Write a php program to display given series

1, 11, 111, 1111, 11111,

1 3 7 13 . . 10 th term

```
<?php
     for($i = 1; $i<=5; $i++)
     {
      for($j = 1; $j<5; $++):
    echo "$i";
<?php
$a = 0
 for($i = 1; $i<=5; $i++)
    {
     $a = $ax10 +1
     echo "$a" ;
    }
   ?>
```

```
for(
{
$a = ($i x $i) - $i - 1 }
echo "$a";
}
```

Array

Array is collection of variables having same names in php variables are not type specific that means variables can hold different data type

variables of array differentiate among each other with the help of index value.

Syntax $<array-name> = array(<size> or <data>);

eg

```
$a = array(10);
$b = array(10,20,30,40,50);
```

Types of Array

1. Indexed Array
2. Associative Array (Named Array)

① Index Array

When variables of array identified among each other with the help of indexed number then it is called index array indexed always begins with zero

Write a php program to display salary of ten person using array.

```
<HTML>
<Head>..</Head>
<body>
<?php
$sal = array(20000,30000,35000,25000,40000,15000
            18000,45000,10000,5000)
for($i=0; $i<10; $i++)
{
echo "$sal <Br>";
}
?>
</Body>
</HTML>
```

2. Associative Array

When variables of array identified among eachother with the help of indexed name then it is called associative array. In this array data are stored in <key => value> pair where key is index name

Write a php program to demonstrative associative array

```
<HTML>
<Head></Head>
<Body>
<?php
$a = array("roll"=>101 "name"=>"Ram", "total"=>430);
echo $a["roll"]."<BR>";
echo $a["name"]."<BR>";
echo $a["total"]."<BR>";
?>
```

```
for each ($a as $x)
 {
 echo "$x <BR>"
 }
```

```
for each ($a as $x => $y)
 {
 each "$x = $y <BR>";
 }
```

Function

Function is a block of program code with some name the main purpose of function is to represent a large problem in group of small problems when a program contains a block of program code frequently use then block is defined as a function.

syntax

```
function <function-name>([parameter/arguments])
 {
 }
```

Types of Function (based on defination)

1 pre defination
2 user-defined fuction

(base on structure)

1 function with no return type & no parameter
2 function with no return type but parameter.
3 function with return type and with parameter.
4 function with return type but no parameter.

Write a program to find simple interest using function.

Predefined Global variable

There are some predefined globle variables available in php They are used to transfer data between webpages They are in associative array in nature They are ~~multiple~~ multidimensional array some commonly used global variables are.

- → $ - POST
- → $ - GET
- → $ - SESSION
- → $ - FILES
- → $ - SERVER
- → $ - COOKIES

$_POST is a global variable that is used to transfer data submitted by form. using post method

$_GET is a global variable that is used to transfer data submitted by form using GET method.

$_SESSION is global variable that is used to store data set by user. This data can accessed from multiple / all web pages of the website. It is used to maintain session of user. To Save data in session variable we used session_start();
Similarly to destory session data we use session_destroy();

$_SERVER is a global variable that is used to store Information related to Server and their activities.

$_COOKIES is a global variable that is used to store information related to user transaction or visit history in user computer

Write a program to insert p t r in a form and display simple interest and amount in another Web page

entry.php

```
<HTML>
<HEAD> - - - </HEAD>
<Body>
<H2> simple interest calculator </H2>
<form method = "post" action="display.php">
principle <input type = "text "name = "principle"/><BR>
Time <input type = "text" name = "time"/><BR>
Rate <input type= "text" name = " rate/> <BR>

<input type = "submit "value = "Add "name = "add"/>

</Form>
</Body>
</HTML>
```

display php.p

```
<?php
$p = $-post["principle"];
$t = $-post["time"];
$r = $-post["rate"];

$i = $p*$t*$r/100;
$a = $p+$i;

echo "intrest = $i<BR>";
echo "Amount = $a<BR>";
```

write an php program to enter product detail (id name, price) using form and display the detail along with discount and net amount (criteria for discount ; 10% if price is 5000 otherwise 5%)

entry.php

```
<HTML>
<Head>--- </Head>
<Body>
<H2> Product Details </H2>
<Form method = "post" action = "display.php">
name <input type = "text" name = "name"/> <BR>
id <input type = "text" name = "id" /> <BR>
price <input type = "text" name = "price" /> <BR>
<input type = "submit" value = "Add" name = "add">
</form>
```

display.php

```
<?php
$id = $_post["id"];
$N = $_post["name"];
$p = $-post["rate"];

if ($p >5000)
{
$d = $p* 10/100; $np = $p-$d
echo "Discount = $d <BR>";
echo netprice = $np <BR>;
else
{
$d = $p x 5/100;
$np = $p-$d;
echo "Discount = $d <BR>";
echo "Netprice = $np <BR>";
}
?>
```

write a php program to demonstrate login process and maintain session sing session variables

login.php

```
<HTML>
<HEAD> - - </HEAD>
<Body>
<H1> login form </H1>
<form method = "post" action= "login.php">
Email <input type = "text" name = email "/><BR>
password <input type= "password" name = pwd"/> <BR>

<input type = "Submit" value = "login" name = "Login"/>
</form>
</Body>
</HTML>
<?php
if (isset($_POST["login"]))
{
$e = $_post["email"];
$p = $_post["pwd"];
if ($e == "ram@gmail.com" && $p = "123")
{
session_start();
$_SESSION ["email"] = $e;
header ("Location: dashboard.php");
}
else
{
echo "login Error";
}
}
?>
```

dashboard.php

```
<?php
session_start();
if (!isset ($_SESSION["Email"]))
{
header ("location: login.php");
}
?>
<HTML>
<Head> - - </Head>
<Body>
<H1> welcome to Dash board </H1>
<?php echo $_SESSION ["Email"]; ?>
</Body>
</HTML>
```

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Database connectivity

To work with database whe we have to follow the procedures provided by DBMS DBMS is seperate application so we require some information to work with database information like location of database user name password & database name. To work with database we have to following steps:

step1
connect with database (in php/mysql we use we we mysql-connect () or new mysqli() constructor) where server name, user name password & database name is provided as a parameter.

step 2:
verifiy connection (with the help of connect-error flag we can verify connection if it is true then there is error other wise success.

step3: write SQL command
Step4: Execute Query
Step5: Display result

Q.N Write the ~~P~~ php program to create a data base ~~db-pr~~ ~~d in my sql~~
db-product in mysql server

$\boxed{\text{Create Db. php}}$

```
<?php
$conn = new mysqli("lochost", "root", " ");
if($conn -> connect - error)
{
      die("connection error");
}
$sql = "create database db-product";
$r=$ conn-> query ($sql);
if($r)
 {
 echo "Database created";
 }
else{
     echo "Error";
 }
?>
```

error handling

$\boxed{\text{Further execution stop}}$

Q.N Write a php program to create a product table (Id, name, category, price) in Db- product data base

Create table.php

```
<?php
$conn = new mysqli("lochost", "root", " ","db-product");
if ($conn -> connect _error)
{
 die (" connection Error"); }
$sql = "create table product(id int primary key,name char(50), category char (50) price, int]";

$r = $ conn -> query ($sql);
   if ($r)
   {
    echo "table created";
   }
   else
    {
    echo "error";
    }
    ?>
```

Write a php program to insert a product record in product table

```
create table php
<?php
$conn = new mysqli("lochost","root", "" "db_product");
if ($conn -> connect - error)
{
die("connection Error");
}
$sql = "insert into product values ('101', 'computer' 'electsonics', '50000')";
$r = $conn -> query ($sql);
    if ($r)
    { echo "Record Inserted";
    }
    else {
        ~~efor~~ echo "error";
      }
    ?>
```

Q. Write a php program to update product price set 60000 of product id 101.

```
create table.php
<?php
$conn = new my sqli ("lochoct","root", " ""db-product");
if ($conn -> connect - error)
{
die ("connection error");
}
$ sql = " update product set price = '60000' where id = '101' ";
$r  = $conn -> query ($sql);
    if ($r)
   { echo " Record updated ";
    }
     else{
        echo "error";
       }
       ?>
```

Q.1 Write a php program to delete product record of product id 101

createtable.php

```
<?php
$conn = new mysqli("localhost", "root", "", "db_product");
if($conn -> connect_error)
{
  die("connection error");
}
$sql = "delete from product where id='101'";
$r = $conn -> query($sql);
if($r)
{
  echo "Record Deleted";
}
else
{
  echo "Error";
}
?>
```

Q.1 Write a php program to display record product table

createtable.php

```
<?php
$conn = new mysqli("loshost", "root", "", "db-product");
if($conn -> connect_error)
{
  die("connection error");
}
$sql = "select * from product";
$r = $conn -> query($sql);
while($row = $r -> fetch_assoc())
{
  $i = $row["id"];
  $n = $row("name");
  $c = $row("category");
  $p = $row("price");

echo "ID = $i <BR> Name = $n <BR>
category = $c <BR> price = $p <BR><BR>";

}
?>
```

Qn Write a php program to enter product detail (id, name, price) in a form & store it in product table

Entry.php

```
<HTML>
<Head> - - - </head>
<body>
<h1> product entry form </h1>
<form method ="post" action = "insert.php">

Id <input type ="text" name = "id" /> <BR>
name <input type = "text" name = "name" /> <BR>
price <input type = "text" name = "price" /> <BR>

</form>
</body>
</HTML>
```

insert.php

```
<?php
if (isset ($_post ["insert"] ))
{
  $i = $_post ["id"];
  $n = $_post ["name"];
  $p = $_post ["       "];

 $conn = new mysqli ("local host", "root", "", ~~db_product~~
  "db_product");
 if ($conn -> connect_error )
  {
    die(" Connectors_Error ");
  }
 $sql = " insert into product values ('$i', '$n',
 '$p' )";

 $r = $conn -> query ($sql);
  if($r){
   echo " Data inserted ";
  }
  else
  {
   echo "Error";
  }
 }
?>
```

Write a php program to create a registration form (Name Email, password) & store the detail in data base.

register.php

```
<HTML>
<Head> - — - - </Head>
<body>
<h2> Registration from </h2>
<from method= "post" action = "register.php".
name <input type= "text" name = name> <Br>
Email <input type = "text" name= Email <BR>
password <input type = "password" name= "
password" /> <BR>
input type = "text value= "submit" name=
"submit" />
~~</body>~~
</from>
</body>
</HTML>
<?php
if(isset($_post["submit"])
{ $n = $_post ["id"];
  $e = $_post ["email"];
  $p = $_post ["password"];

$conn= new mysqli ("local host", "root", "",
   db_product");
 if($conn -> connect_error)
{
 die("connection Error");
}
$sql = "insert into register value ('$n', '$e', '$p')";

$r = $conn -> query ($sql);
  if ($r) {
   echo "Data inserted";
  }
 else
  {
  echo "Error"; }
  }
  } ?>
```

DML

Insert command

It is used to insert a record in the table.

syntax:

Insert into <table name> ([field_name])
values('<value,>', '' '....', ...);

→ insert into user (name, email, password)
values ('Subash', subash@gmail.com', '123');

→ insert into user values ('0', 'suman',
suman@gmail.com', '123' '9822334455');

update command

It is used to update is used to change the value of records.

syntax

update <table name>
set <field_name] = [new value]: ----
where [condition];

eg update user
set password = 'abc' mobile = '989....
where id = '2';

Delet command

It is used to delet records from the table,

syntax

delete form <table_name >
where < condition >;

eq: delete from user
where id = '2'

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Q.) what do you mean by well formed xml document & valid XML document,

→ when XML document follow syntax rules & naming rules of XML document then document is consider as well-formed document

→ similarly, when XML document is validated using schema document either DTD (document type defination or XML schema) then the document is called validated XML document,

Q.) Demostrate with example? XML naming rules & syntax rule

* DTD

It is a schema language that is used to define structure of XML Document & enforce the data stored in XML document must be place according to DTD definations / rules

Type of DTD defination

1) Internal DTD

2) External DTD

1) Internal DTD

When DTD codes are defined inside XML document then it is called internal DTD

2 External DTD → when DTD codes are define in a seperate file with externtension. dtd & then the file is liked to the XML document then it is called external DTD. External DTD may defined public or private. If it is defined private the the DTD file can be liked locally to the XML document with in the same website or server. If it is defined public then the DTD file can be linked from any website in www.

Eg

```
<? XML version = "1.0"?>
<! DOCTYPE product [<!ELEMENT products(
    product)*>
  <!ELEMENT product (id, name, price)>
  <!ELEMENT id(#PCDATA)>
  <!ELEMENT name(#PCDATA)>
  <!ELEMENT price(#PCDATA)>
  ]>
<products>
  <product>
    <id> 101 </id>
    <name> subigya kumari</name>
    <price> .. .. </price>
    </product>
    :
    :
<products>
```

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Create an XML document that stored data of products (id product_name, category, price size color) size category & color is optional element define DTD file as well.

Product.dtd

```
<! ELEMENT product ( product )* >
<! ELEMENT product (id, name, category*. price, size*
  color* )>

<! ELEMENT id [# pcDATA]>
<! ELEMENT name [# pcDATA]>
<! ELEMENT category [# pcDATA]>
<! ELEMENT price [# pcDATA]>
<! ELEMENT size [# pcDATA]>
```

Product.XML

```
<? XML version = "1.0"?>
<! DOCTYPE product SYSTEM "product.dtd">
<product>
      <id>101<id>
      <name>tap</name>
      <category> electronic </category>
      <price >2500</price>
      <color> yellow </color>
      </product >
<products>
<id > 102 </id >
<name> cloath</name>
<price> 1500</price>
</products></products>
```

Define DTD File For given XML document

```
<?xml version = "1.0"?>
<!DOCType accounts System "acc.dtd">
<accounts>
  <account type = "saving" num = "101">
  <name> </name>
  <adress> </Address>
  <balance> </balance>
  <mobile> </mobile>
  <branch> </branch>
  </account>
  <account>
      :
<accounts>
```

Note: numbers should be unique type can be saving current fixed address can repeated.

acc.dtd

```
<!ELEMENT accounts(account)*>
<!ELEMENT accounts(name, address balance, mobile
                     branch)>
<!ELEMENT name (#PCDATA)
<!ELEMENT address   ,,
<!ELEMENT balance   ,,
<!ELEMENT mobile    ,,
<!ELEMENT blance    ,,
<!ATTLIST account type ("Saving" |current" |"Fixed")>
<!ATTLIST account num ID>
```

XML schema

XML schema is a schema language that is used to defined the ~~str~~ structure of XML document It is alternative & advance to the document type defination It support data type & name space It is also refered to as XML schema Definition (XSD) It is also used to validate XML document.

write XML schema code to define XML schema ~~for~~ product document product

product.XML

```
<?xml version = "1.0"?>
  <product>
    <id> </id>
    <name> </name>
    <category> </category>
    <price> </price>
  </product>
```

product.xsd

```
<?xml version = "1.0"?>
<xs: schema>
<xs: element name = "product">
```

```
<xs: complex Type
<xs: sequence >
    <xs: element name = "id" type = "xs :integer"/>
    <xs: element name = "name" type= "xs: string"/>
    <xs: element name = " category" type = "xs: string"/>
    <xs: element name = " price" type= "xs: integer"/>
    <xs: element

<lxs: sequence>
<lxs: complex Type>
</xs: element >
</xs: schema>
```

2022-1-25

write XML Schema code for given XML document use appropriate restriction.

```
<?XML version ="1.0" ?>
<studentes>
 <student>
   <roll>1</roll>        // 1to 70 roll
   <name> --- </name>
   <address> --- </address>
   <program> --- </program>    BCA / B.ed / MA
   <duration> --- </duration>    2,2r
   <sem> --- </sem> (1 to 8)
 <student>
   |
   |
 </student>

<?XML version="1.0" ?>
<xs: schema>
<xs: Element name = "student" >
<xs: ~~Element~~ complex Type
   <xs: element name = " student" >
   <xs: complex type >
     <xs: sequence >
     <xs: element name = "roll" >
     <xs: simple Type
     <xs: restriction base = "xs : string" >
     <xs: minlength value="1"/>
     <xs: maxlength value = " 70"/>
     <xs: restriction ~~base~~
     <xs: simple Type >
```

```
<xs: element>
<xs: element name "address" type=" xs: string">

<xs: element name "program" ~~type="xs:string"~~ >
<xs: simpleType>
<xs: restriction base = "xs: string")
<xs: enumaration value = "BCA"/>
<xs: enumeration value = "BA"

</xs: restriction >
</xs: simpleType>
</xs: element >
<xs: element name "duration">
<xs: simple Type >
<xs: restriction base = "xs: integer">
<xs: mininclusive value = "2"
<xs: max inclution value = "8"

</xs: restriction >
</xs: simple Type>
</xs: element>

<xs: element name "Sem">
<xs: simple Type>
<xs: restriction base = "xs: integer",
<xs: mininclusive value = "1">
<xs: max inclusive value = "8">
</xs: restriction>
</xs, simple type>
</xs; element>
```

```
</xs: sequence >
</xs: complexType>
</xs: element type>
</xs: complex Type>
</xs: element>
</xs: schema
```

write XSD code to define scheme for given XML document

```
<?xml version = "1.0" ?>
<products>
  <product id = "101">
    <name> --- </name>
    <color name = "read" />
 <category> ------ </category>
 <price discount = "10"> --- </price>
 <discription>
      ------
         <mfd_date> --- </mfd_date> --
         ~~</product>~~        </discription>
         </product>
          .
          .
          .
<products>
```

```
<?xml version="1.0"?>
~~product~~
<xs: schema>
<xs: element name = "products">
<xs: coplex Type
   <xs: Element name = "product">
   <xs: compler Type><xs; attribute id = "101">
   <xs: ~~element name~~ sequen(e>
   <xs: element name = "name" type = "xs:string">

  <xs: element name = "color">
     <xs: complexType>
     <xs: attribute name = "red" typ = xs:string">
    </xs: complex Type>
    </xs; element >
 <xs: element name = "category" type = "xs:string">
 <xs: element name = "price">
 <xs: compley Type >
<xs: simple ~~Type~~ content
<xs: attribute name = " discount " type = "xs:integer
<xs: simple content>                     ">
</xs: complex Type>
</xs: element >
<xs element name = "description">
<xs: complex Type mixed = "true">
<xs: element name = "mfd_date"> type
          = "xs: date">

<xs: complexType>
<xs: element>
<xs: attribute name = "id" type = "xs:integer"
        use = "required">
</xs: sequence>
</xs: complexType>
</xs: element>
</xs: complexType>
</xs: element>
</xs: schema>
```

Indicator

~~define an xml schema~~

~~write~~ a XML code and it's schema to store following information about student

each student has a name, address phone and website element

Address might appears multiple time

Address has attributes name "type" with value permanet and temporary

phone must be 10 digit

```
<xml version "1.0"?>
<Students>
<student >
<name > ---- </name>
<address> — — </Address>
<phone> — — </phone>
< webside element > - - - -          </ weside element
</ Student>
</Students>
```

```
<xML version "1.0" ?>

<xs: schama>
<xs: element name = "students'>
<xs: complex Type
  <xs: element name = "student'>
  <xs: complex Type
  <xs: Sequence>

  <xs: element name = "name" type = "xs:string">
  ~~<xs:~~
<xs: element name = "Address" Type = "minoccurure=
       "1"  max occurure "2" >

<xs: complex type
<xs: attribute  name = "permoned" type = "xs: string">
<xs: attribute name = "temporary" type = "xs: string">
</xs: complex Type>
</xs: element>
<xs: element  name: "phone" Type = "xs: integer">

<xs : element name: "webside" Type = "xs: string" >
~~<xs : complex Type>~~
</xs: Sequence >
</xs: complex type >
</xs: element >
</xs: Complex Type>
</xs: element>
</xs: Schama>
```

XQuery

What write xQuery expression to display employee details (Id name post salary) where salary is greter the 2500 the date of emlopee is store in employee xml

```
<?xml version "1.0" ?>
<employee>
<employee>
<id> --- </id>
<name> --- </name>
<post> --- </post>
<salary> --- </salary>
</employee>
:
</employee>
```

```
<HTML>
<Body>
<H1>employee Detail </H1>
let $emp = doc("employee.xml")/employee/employee

return
{
for $e in $emp
where $e/salary >= 2000
return {
data($e/id) } <BR>
{ data($e/name) } <BR>
{ data($e/post) } <BR>
{ data($e/salary) } <BR>
```

Alternative अन्य तरीका

```
for $e in doc("employee.xml")/employees/employee
where $e/salary >= 2500
return { data($e/id) <BR>
data($e/name post id>
```